

Responsibility

Primary: Data Structures & Process Management

Supporting: Testing

1. Process Data Structure

Each process will contain:

- Process ID
- Process State
- Program Counter
- Stack Pointer
- Priority

2. Process States

A process can have the following states:

New
Ready
Running
Waiting
Terminated

3. Process Management

The simulator will maintain a collection of processes.

Each process will have a unique Process ID.

The process manager will be responsible for:

- Creating processes
- Maintaining process information
- Changing process states
- Removing terminated processes

4. Initial Testing Plan

The following will be tested during implementation:

1. Process creation
2. Process state transition
3. Process ID handling
4. Process selection
5. Empty process list handling